[實驗名稱]

組別

王翊權 E24146107

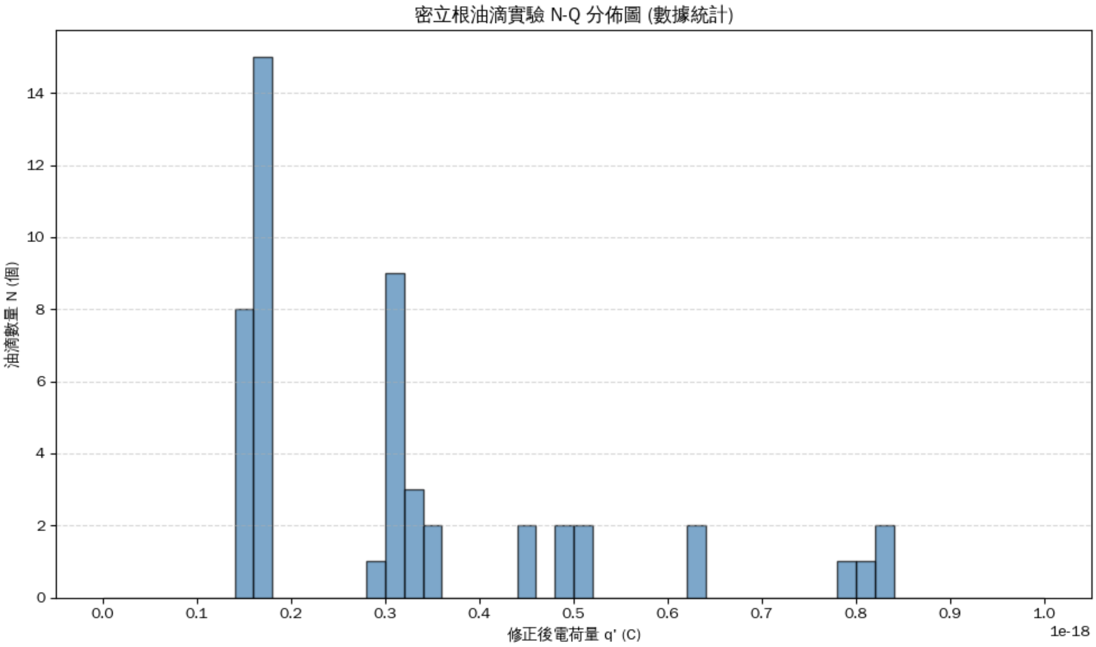
一、實驗數據

次數	X(格數)	S(m)	T(s)	V1(m/s)	U0(V)	Q(coulomb)
1	1	0.000382	19.71	1.9381E-05	102.5	1.61905E-19
2	1	0.000382	10.13	3.77098E-05	93.4	4.82229E-19
3	1	0.000382	25.52	1.49687E-05	73.1	1.54091E-19
4	1	0.000382	17.53	2.17912E-05	115.8	1.70857E-19
5	1	0.000382	23.62	1.61727E-05	78.4	1.61354E-19
6	1	0.000382	21.06	1.81387E-05	47.3	3.17663E-19
7	1	0.000382	7.22	5.29086E-05	118.2	6.33274E-19
8	1	0.000382	18.28	2.08972E-05	42.1	4.41335E-19
9	1	0.000382	10.24	3.73047E-05	141.5	3.1319E-19
10	1	0.000382	10.35	3.69082E-05	268.4	1.62488E-19
11	1	0.000382	19.62	1.94699E-05	54.2	3.08295E-19
12	1	0.000382	9.82	3.89002E-05	148.6	3.17562E-19
13	1	0.000382	21.77	1.75471E-05	93.6	1.52739E-19
14	1	0.000382	10.27	3.71957E-05	272.5	1.61917E-19
15	1	0.000382	11.16	3.42294E-05	224.8	1.73269E-19
16	1	0.000382	11.09	3.44454E-05	125.1	3.14311E-19
17	1	0.000382	11.04	3.46014E-05	76.5	5.17487E-19
18	1	0.000382	14.19	2.69204E-05	158.4	1.71509E-19
19	1	0.000382	16.72	2.28469E-05	143.1	1.4843E-19
20	1	0.000382	7.52	5.07979E-05	137.8	5.11021E-19
21	1	0.000382	14.71	2.59687E-05	167.4	1.53759E-19
22	1	0.000382	24.8	1.54032E-05	73.9	1.59108E-19
23	1	0.000382	19.74	1.93516E-05	102.4	1.61694E-19
24	1	0.000382	11.06	3.45389E-05	124.4	3.17367E-19
25	1	0.000382	17.2	2.22093E-05	125.6	1.62081E-19
26	1	0.000382	5.45	7.00917E-05	141.2	8.08323E-19
27	1	0.000382	24.13	1.58309E-05	41.1	2.98083E-19
28	1	0.000382	16.18	2.36094E-05	131.8	1.6929E-19
29	1	0.000382	22.61	1.68952E-05	86.8	1.55612E-19

30	1	0.000382	17.65	2.16431E-05	61.4	3.18955E-19
31	1	0.000382	14.73	2.59335E-05	156.4	1.64238E-19
32	1	0.000382	21.79	1.7531E-05	89.2	1.60053E-19
33	1	0.000382	16.9	2.26036E-05	138.8	1.5059E-19
34	1	0.000382	12.78	2.98905E-05	64.8	4.90505E-19
35	1	0.000382	10.06	3.79722E-05	276.4	1.64657E-19
36	1	0.000382	10.27	3.71957E-05	55.2	7.99319E-19
37	1	0.000382	6.63	5.76169E-05	256.1	3.3215E-19
38	1	0.000382	9.93	3.84693E-05	270.2	1.71753E-19
39	1	0.000382	18.13	2.107E-05	58.7	3.20464E-19
40	1	0.000382	23.57	1.6207E-05	80.6	1.57449E-19
41	1	0.000382	14.91	2.56204E-05	81.3	3.10247E-19
42	1	0.000382	7.26	5.26171E-05	230.7	3.21782E-19
43	1	0.000382	15.32	2.49347E-05	75.8	3.1949E-19
44	1	0.000382	10.75	3.55349E-05	49.5	8.32332E-19
45	1	0.000382	16.17	2.3624E-05	137.9	1.61952E-19
46	1	0.000382	6.35	6.01575E-05	262.4	3.45852E-19
47	1	0.000382	11.1	3.44144E-05	63.3	6.20335E-19
48	1	0.000382	17.78	2.14848E-05	43.5	4.45274E-19
49	1	0.000382	11.82	3.23181E-05	43.1	8.29108E-19
50	1	0.000382	18.44	2.07158E-05	53	3.46017E-19

二、數據分析

區間	個數
1.00E-20	0
2.00E-20	0
3.00E-20	0
4.00E-20	0
5.00E-20	0
6.00E-20	0
7.00E-20	0
8.00E-20	0
9.00E-20	0
1.00E-19	0
1.10E-19	0
1.20E-19	0
1.30E-19	0
1.40E-19	0
1.50E-19	1
1.60E-19	7
1.70E-19	11
1.80E-19	4
1.90E-19	0
2.00E-19	0
2.10E-19	0
2.20E-19	0
2.30E-19	0
2.40E-19	0
2.50E-19	0
2.60E-19	0
2.70E-19	0
2.80E-19	0
2.90E-19	0
3.00E-19	1
3.10E-19	1
3.20E-19	8
3.30E-19	2
3.40E-19	1
3.50E-19	2
3.60E-19	0
3.70E-19	0
3.80E-19	0
3.90E-19	0
4.00E-19	0
4.10E-19	0
4.20E-19	0
4.30E-19	0
4.40E-19	0
4.50E-19	2
4.60E-19	0
4.70E-19	0
4.80E-19	0
4.90E-19	1
5.00E-19	1
大於5.0E-19	8



https://colab.research.google.com/drive/1u7ZSYuyIYGatZY50uCPZ9nbXALKb7kl_?usp=sharing

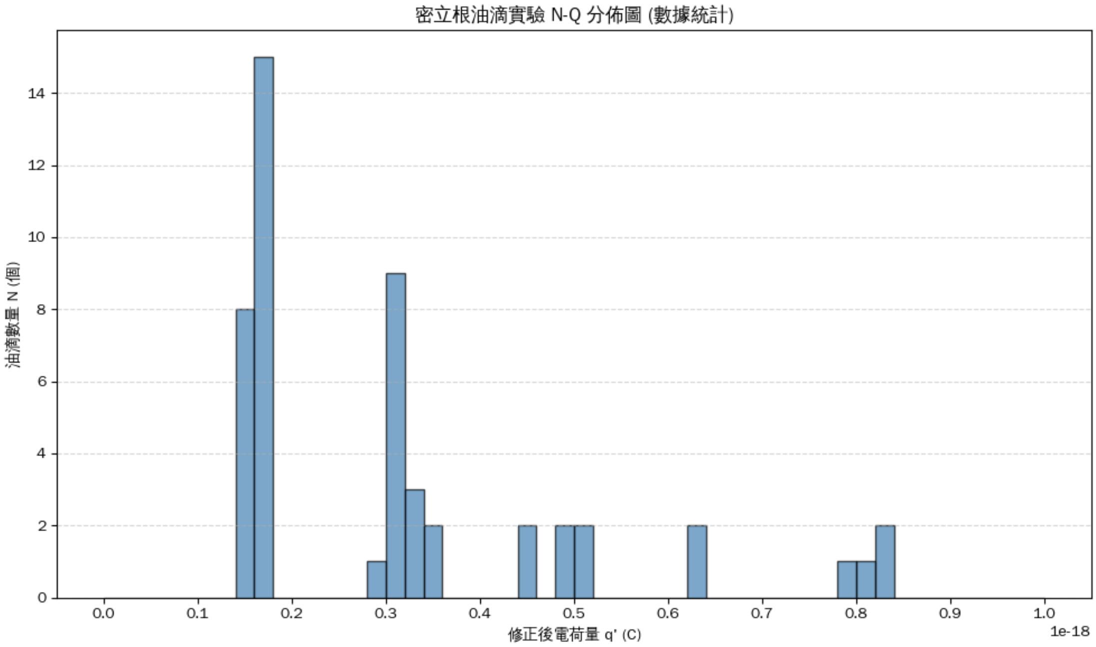
由 50 組實驗觀測值計算出修正電荷量 q'。將各油滴的q'除以對應的 n，求得基本電荷測量值

$$e_{exp} = 1.608 \times 10^{-19}\,\mathrm{C}$$

與基本電荷理論值 $1.602 \times 10^{-19}\,\mathrm{C}$ 相比，百分誤差為 0.39%。實驗數據繪製成 N-Q 圖後呈現明顯的群聚分佈特徵。油滴的電荷量皆集中於 $1.6 \times 10^{-19}\,\mathrm{C}$ 的整數倍區間，驗證了電荷的量子化。

三、誤差來源與解釋

操作碼表記錄油滴通過顯微鏡刻度線時會因為視覺與動作反應延遲，使 t_f t_r產生誤差。平行極板邊緣存在電場不均勻現象，且直流電源供應器輸出電壓存在微小波動，導致油滴受到的靜電力沒有恆定。



密立根油滴實驗 N-Q 分佈圖 (數據統計)
油滴數量 N (個)
0
2
4
6
8
10
12
14
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1e-18
修正後電荷量 q' (C)